Prof. Dr. Sebastian Höhna

Summary

Funding	21 grants/fellowships in 5 countries total ≈ 5.0 Mio €	Supervision	8 PhD students (2 finished) 6 postdoctoral researcher
Publications	58 Journal articles 4 preprints 1 conference proceeding article	Research profiles	Google Scholar Web of Science Orcid ID
Citations	34,544 (Google Scholar) 18,156 (Web of Science)	H-Index	29 (Google Scholar) 22 (Web of Science)

Academic Positions

May 2024-present	W2 Professor, Ludwig-Maximilians-Universität, München, Germany
Mar 2020-Dec 2020	50% parental leave (Elternzeit)
Jan 2019-Apr 2024	Emmy Noether Research Group Leader, Ludwig-Maximilians-Universität, München, Germany
Apr 2017-Dec 2018	Research Group Leader (Akademischer Rat), Ludwig-Maximilians-Universität, München, Germany

Education

Sept 2014-Mar 2017	Miller fellow (Postdoc), University of California, Berkeley, USA Host: John P. Huelsenbeck
Jan 2014-Aug 2014	Postdoc, University of California, Davis, USA Supervisor: Brian R. Moore
Apr 2009-Nov 2013	PhD in Mathematical Statistics, Stockholm University, Sweden PhD Thesis defended on 29th November 2013 Licenciate Thesis presented in December 2011 Supervisors: Tom Britton and Fredrik Ronquist
Oct 2012-Mar 2013	Erasmus Exchange Semester in Mathematics, Technical University Munich, Germany
Mar 2008-Feb 2009	Master of Science in Computer Science, The University of Auckland, New Zealand with First Class Honors Supervisor: Alexei J. Drummond
Mar 2007-Feb 2008	Postgraduate Diploma in Computer Science, The University of Auckland, New Zealand with Distinction
Sept 2003-Aug 2006	Diplom-Engineer in Computer Science, Berlin School of Economics and Law (Berufsakademie Berlin), Germany

Work Experience

Sept 2006 - Jan 2007	Systems Engineer at Bayer-Schering-Pharma AG, Berlin, Germany
Jul 2005 - Sept 2005	Software Engineer at Nihon Schering KK, Osaka, Japan
Sept 2003 - Aug 2006	Software Engineer at Schering AG, Berlin, Germany

Teaching Experience

University Classes:

2025/6	*Data Analysis in Geobiology and Paleontology*	Lecturer	LMU München, Germany
2025/6	*Individual Research Projects (Master MGAP)*	Coordinator	LMU München, Germany
2025	*Phylogenetics & Macroevolution*	Lecturer	LMU München, Germany
2025	*Statistics and Data Analysis*	Lecturer	LMU München, Germany
2025	*Scientific Presentation and Communication*	Guest Lecturer	LMU München, Germany
2024/5	*Phylogenetic Analysis of Morphological and Molecular Characters*	Lecturer	LMU München, Germany
2024	*Scientific Presentation and Communication*	Guest Lecturer	LMU München, Germany
2022	*Phylogenetics & Macroevolution*	lectures and practicals	LMU München, Germany
2019	*Programming for Bayesian phylogenetics*	lectures and practicals	LMU München, Germany
2018	*Computational Phylogenetics*	practicals	LMU München, Germany
2018	*Evolutionsbiologie*	lectures and practicals	LMU München, Germany
2018	*Phylogenetics & Macroevolution*	lectures and practicals	LMU München, Germany
2017	*Computational Phylogenetics*	practicals	LMU München, Germany
2017	*Evolutionsbiologie*	lectures and practicals	LMU München, Germany
2016	*Phylogeny and Macroevolution*	Guest Lecturer	UC Davis, California, USA
2011	*Bayesian Methods*	lectures and practicals	Stockholm University, Sweden
2009	*Mastering Cyberspace: An introduction to practical computing*	lectures	The University of Auckland, New Zealand
2008	*Computational Biology*	practicals	The University of Auckland, New Zealand
2007	*Algorithms and data structures*	practicals	The University of Auckland, New Zealand

International Workshops:

Oct 2023	*Bayesian inference of phylogeny with tip-dating Workshop*	Munich, Germany
Nov 2022	*Total-Evidence with RevBayes Workshop*	Oslo, Norway
Apr 2021	*Stay-at-Home RevBayes Workshop Spring 2021*	online
Feb 2021	*Mathematical, Inferential, and Computational Phylogenomics*	online
Sept 2019	*Bayesian phylogenetics and macroevolution in RevBayes*	LMU München, Germany
July 2019	*Phylogenomics and Population Genomics: Inference and Applications*	University of Barcelona, Spain
June 2019	*MadPhylo: Madrid Workshop on Phylogenetics*	Royal Botanical Garden, Madrid, Spain
May 2019	*Applied Phylogenetics Workshop*	Bodega Bay, California, USA
Dec 2018	*Analysing Macroevolutionary Processes using RevBayes*	Université de Montpellier, France
Aug 2018	*MadPhylo: Madrid Workshop on Phylogenetics*	Royal Botanical Garden, Madrid, Spain
May 2018	*Analysing Macroevolutionary Processes using RevBayes*	University of Bristol, UK
Apr 2018	*Estimating time-calibrated phylogenies and diversification rates*	MACN, Buenos Aires, Argentina
Aug 2017	*Bayesian phylogenetics using RevBayes*	NIMBioS, Knoxville, Tennessee, USA
Mar 2017	*Applied Phylogenetics Workshop*	Bodega Bay, California, USA
Jan 2017	*Introduction to Bayesian Phylogenetic inference using RevBayes*	SSB Meeting, Baton Rouge, Louisiana, USA
July 2016	*Bayesian Phylogenetic inference using RevBayes*	UC Davis, California, USA
Mar 2016	*Bayesian Phylogenetic inference using RevBayes*	UC Irvine, California, USA
Sept 2015	*Bayesian Phylogenetic inference using RevBayes*	LSU, Louisiana, USA
Apr 2015	*Bayesian Phylogenetic inference using RevBayes*	UC Berkeley, California, USA
Mar 2015	*Applied Phylogenetics Workshop*	Bodega Bay, California, USA
Feb 2015	*Molecular Evolution Workshop*	Cesky Krumlov, Czech Republic
Aug 2014	*Bayesian Phylogenetic inference using RevBayes*	NESCent, Durham, NC, USA
Mar 2013	*Bayesian Phylogenetic inference using RevBayes*	University of Groningen, The Netherlands
Aug 2012	*Virus Evolution and Molecular Epidemics*	Belgrade, Serbia

Student & Postdoc Supervision

Sep 2026 – Present	Chiara Zanoli, PhD student
Aug 2025 – Present	David Černý, Postdoc
May 2025 – Present	Priscilla Lau, PhD student
May 2024 – Present	Wenjie Zhu, PhD student
Feb 2024 – Present	Haoqing Du, PhD student
Sep 2023 – Present	Basanta Khakurel, PhD student
Sep 2023 – Present	Gustavo Darlim, Postdoc
Jul 2022 – Present	Alessio Capobianco, Postdoc
Feb 2020 – Jan 2026	Bjørn Tore Kopperud, PhD student (graduated in Jan 2026)
Nov 2019 – Sep 2024	Ronja Jessica Billenstein, PhD student
Feb 2023 – Sep 2024	Nicola Heckeberg, Postdoc
May 2022 – Feb 2024	John Clarke, Postdoc (Humboldt Fellowship)
Nov 2019 – Dec 2022	Killian Smith, PhD student (now a research programmer at HPCNow!)
Mar 2019 – Oct 2022	Luiza Fabreti, PhD student (graduated in April 2023; now a research programmer at camLine)
Mar 2019 – Feb 2021	Allison Hsiang, Postdoc (now independent researcher at Stockholm University)
2009 – Present	supervision of several undergraduate and master-level student projects

Visiting Students & Postdocs

Sep 2025 – Dec 2026	Junru Zhang, PhD Student, Northwest A&F University, China, CSC fellowship
Nov 2025 – Feb 2026	Angelá Aguado-Lara, PhD Student, Real Jardín Botánico (CSIC), Madrid, Spain, Comunidad de Madrid fellowship
Nov 2024 – Mar 2025	Fernanda Landim, MSc Student, University of So Paulo (USP), Brazil, FABESP Exchange
cancelled due to COVID19	Ignacio Quintero, Postdoc, Marie Curie Secondment, primary host: Hélène Morlon, Ecole Normale Supérieure in Paris
Sept 2019 – Oct 2019	Andrew F. Magee, PhD Student, University of Washington, Seattle, DAAD Fellowship

Publications

Submitted preprints:

[S1] **Höhna S**, Du H and Catalán A: Impact of Partition Models on Phylogenetic Inference and Divergence Times of Lampyridae from Mitochondrial Genomes, *bioRxiv*, doi:10.1101/2025.08.19.671050.

[S2] Catalán A, Peréz CMA, Echeverría Méndez AM, Guitiérrez JE, Monzón J, Vega Badillo V and **Höhna S**: Filling the gap in the Neotropics: new firefly species provide insights into Central American biodiversity, *bioRxiv*, doi:10.1101/2025.02.20.639311.

[S3] Darlim G and **Höhna S**: Mixture Models for Dating with Confidence, *bioRxiv*, doi:10.1101/2024.09.25.614964.

[S4] Magee AF, **Höhna S**: Impact of K-Pg Mass Extinction Event on Crocodylomorpha Inferred from Phylogeny of Extinct and Extant Taxa, *bioRxiv*, doi:10.1101/2021.01.14.426715.

Publications in peer reviewed journals:

[J1] Lau P, Kopperud BT, Clarke JT, Metzler D and **Höhna S**: An efficient Bayesian phylogenetic approach for joint inference of continuous and discrete trait evolution under a state-dependent Ornstein-Uhlenbeck model. *Methods in Evolution and Ecology*, 2026, 17:2629–2644, doi:10.1111/2041-210x.70371.

[J2] Capobianco A, Darlim G and **Höhna S**: How to date a molecular phylogeny: comparison of effective priors between node calibration and FBD, *Proceedings of the Royal Society B*, 2026, 292:20260091, doi:10.1098/rspb.2026.0091

[J3] Khakurel B and **Höhna S**: A covarion model for phylogenetic estimation using discrete morphological datasets, *Systematic Biology*, 2026, .doi:10.1093/sysbio/syag014.

[J4] **Höhna S**, Freyman WA, Nolen Z, Huelsenbeck JP, May MR, Rannala B and Moore BR: Inferring branch-specific rates of lineage diversification under the birth-death-shift process, *Systematic Biology*, 2026, 75 (5), 948–966. doi:10.1093/sysbio/syag003.

[J5] Kopperud BT, Capobianco A, Clarke JT, Palazzesi L and **Höhna S**: The nature and prevalence of diversification rate shifts across the Tree of Life, *Evolution Letters*, 2026, 10 (2), 217–227. doi:10.1093/evlett/qrag005.

[J6] Heckeberg NS, Capobianco A, Khakurel B, Darlim G and **Höhna S**. Practical Guide and Review of Fossil Tip-Dating in Phylogenetics. *Systematic Biology*, 2026, 75 (1), 156–192. doi:10.1093/sysbio/syaf050.

[J7] Kopperud BT and **Höhna S**: Phylogenetic Estimation of Branch-specific Shifts in the Tempo of Origination, *Systematic Biology*, 2025, 74 (6), 985–1006, doi:10.1093/sysbio/syaf041.

[J8] Capobianco A and **Höhna S**: On the Mkv Model with Among-Character Rate Variation, *Systematic Biology*, 2025, 74 (6), 952–966, doi:10.1093/sysbio/syaf038.

[J9] [1]Zhu W and **Höhna S**. Two-step Species Tree Inference under the Multispecies Coalescent Using Full-Likelihood. *Evolutionary Journal of the Linnean Society*, 2025, 4 (1), kzaf018, doi:10.1093/evolinnean/kzaf018.

[J10] Paterson RSS, Mackie M, Capobianco A, Heckeberg NS, Fraser D, Munir F, Patramanis I, Ramos-Madrigal J, Liu S, Ramsoee AD, Dickinson MR, Baldreki C, Gilbert M, Sardella R, Bellucci L, Scorrano G, Racimo F, Olsen JV, Penkman KEH, Willerslev E, MacPhee RDE, Rybczynski N, **Höhna S**, Cappellini E: A 20+ Ma old enamel proteome from Canada's High Arctic reveals diversification of Rhinocerotidae in the middle Eocene-Oligocene, *Nature*, 2025, doi: 10.1038/s41586-025-09231-4.

[J11] Catalán A, Gygax D, Candolin U, Tusso S, Duchen P and **Höhna S**: Sex-biased migration and demographic history of the big European firefly Lampyris noctiluca, *Molecular Biology and Evolution*, 2025, 42:6, msaf123, doi:10.1093/molbev/msaf123.

[J12] **Höhna S** and Catalán A: Bayesian StairwayPlot for Inferring Single Population Demographic Histories from Site Frequency Spectra, *Molecular Ecology Resources*, 2025, 25:e14087, doi:10.1111/1755-0998.14087.

[J13] Darlim G and **Höhna S**: The effects of cryptic diversity on diversification dynamics analyses in Crocodylia, *Proceedings of the Royal Society B*, 2025, 292:20250091, doi:10.1098/rspb.2025.0091.

[J14] **Höhna S**, Lower SE, Duchen P and Catalán A: Robustness of divergence time estimation despite gene tree estimation error: a case study of fireflies (Coleoptera: Lampyridae), *Systematic Biology*, 2025, 74 (2), 335–348, doi:10.1093/sysbio/syae065.

[1]Best paper

[J15] Mulvey LPA, May MR, Brown JM, **Höhna S**, Wright AM and Warnock RCM: Assessing the Adequacy of Morphological Models using Posterior Predictive Simulations, *Systematic Biology*, 2025, 74 (1), 34-52, doi:10.1093/sysbio/syae055.

[J16] Khakurel B, Grigsby C, Tran CD, Zariwala J, **Höhna S** and Wright AM: The fundamental role of character coding in Bayesian morphological phylogenetics, *Systematic Biology*, 2024, 73 (5), 861–871, doi:10.1093/sysbio/syae033.

[J17] **Höhna S** and Hsiang AY: Sequential Bayesian Phylogenetic Inference, *Systematic Biology*, 2024, 73 (4), 704–721, doi:10.1093/sysbio/syae020.

[J18] Karapunar B, **Höhna S** and Nützel A: Phylogeny of the longest existing gastropod clade (Pleurotomarida) reconstructed with Bayesian and parsimony methods and its implications on gastropod shell characters, *Journal of Systematic Palaeontology*, 2024, 22(1), 2384141, doi:10.1080/14772019.2024.2384141.

[J19] Billenstein RJ and **Höhna S**: Comparison of Bayesian Coalescent Skyline Plot Models for Inferring Demographic Histories, *Molecular Biology and Evolution*, 2024, 41(5):msae073, doi:10.1093/molbev/msae073.

[J20] Smith K, Ayres DL, Neumaier R, Wörheide G and **Höhna S**: Bayesian Phylogenetic Analysis on multicore Compute Architectures: Implementation and evaluation of BEAGLE in RevBayes with MPI, *Systematic Biology*, 2024, 73 (2), 455–469, doi:10.1101/10.1093/sysbio/syae005.

[J21] Martínez-Gómez J, Song MJ, Tribble CM, Kopperud BT, Freyman WA, **Höhna S**, Specht CD and Rothfels CJ: Commonly used Bayesian diversification-rate models produce biologically meaningful differences on empirical phylogenies, *Evolution Letters*, 2024, 8 (2), 189–199, doi:10.1101/2023.05.17.541228.

[J22] Fabreti L, Coghill LM, Thomson RC, **Höhna S** and Brown JM: The Expected Behaviors of Posterior Predictive Tests and Their Unexpected Interpretation, *Molecular Biology and Evolution*, 2024, 41(3):msae051. doi:10.1093/molbev/msae051.

[J23] Fabreti L and **Höhna S**: Nucleotide Substitution Model Selection Is Not Necessary for Bayesian Inference of Phylogeny With Well-Behaved Priors, *Systematic Biology*, 2023, 72 (6), 1418–1432. doi:10.1093/sysbio/syad041.

[J24] Juravel K, Porras L, **Höhna S**, Pisani D and Wörheide G: Exploring genome gene content and morphological analysis to test recalcitrant nodes in the animal phylogeny, 2023, *PLoS one*, 18 (3): e0282444. doi:10.1371/journal.pone.0282444.

[J25] Kopperud BT, Magee AF and **Höhna S**: Rapidly changing speciation and extinction rates can be inferred in spite of non-identifiability, *Proceedings of the National Academy of Sciences*, 2023, 120 (7), e2208851120, doi:10.1073/pnas.2208851120.

[J26] **Höhna S**, Kopperud BT and Magee AF: CRABS: Congruent Rate Analyses in Birth-death Scenarios, *Methods in Ecology and Evolution*, 2022, 13, 2709–2718, doi:10.1111/2041-210X.13997.

[J27] Catalán A, **Höhna S**, Lower SE, Duchen P: On the inference of a southern origin of the North American firefly *Photinus pyralis*, *Journal of Evolutionary Biology*, 2022, 35, 1488–1499, doi:10.1111/jeb.14094.

[J28] Borges R, Boussau B, **Höhna S**, Pereira RJ and Carolin Kosiol C: Polymorphism-aware estimation of species trees and evolutionary forces from genomic sequences with RevBayes, *Methods in Ecology and Evolution*, 2022, 13, 2339–2346, doi:10.1111/2041-210X.13980.

[J29] Barido-Sottani J, Justison JA, Borges R, Brown JM, Dismukes W, Petrucci BR, Fabreti L, **Höhna S**, Landis MJ, Lewis, PO, May MR, Mendes FK, Pett W, Redelings BD, Tribble CM, Wright AM, Zenil-Ferguson R and Heath TA: Lessons learned from organizing and teaching virtual phylogenetics workshops. *The Bulletin of the Society of Systematic Biologists*, 2022, 1(2):8425, doi:10.18061/bssb.v1i2.8425.

[J30] Szöllősi GJ, **Höhna S**, Williams TA, Schrempf D, Daubin V, Boussau B: Relative time constraints improve molecular dating, *Systematic Biology*, 2022, 71 (4), 797–809, doi:10.1093/sysbio/syab084.

[J31] Palazzesi L, Hidalgo O, Barreda VD, Forest F, and **Höhna S**: The rise of grasslands is linked to atmospheric CO_2 decline in the late paleogene. *Nature Communications*, 2022, 13 (1), 293, 1–10, doi:110.1038/s41467-021-27897-y.

[J32] Tribble CM, Freyman WA, Landis MJ, Lim JY, Barido-Sottani J, Kopperud BT, **Höhna S**, May MR: RevGadgets: an R Package for visualizing Bayesian phylogenetic analyses from RevBayes, *Methods in Ecology and Evolution*, 2022, 13 (2), 314–323, doi:10.1111/2041-210X.13750.

[J33] Fabreti L and **Höhna S**: Convergence assessment for bayesian phylogenetic analysis using mcmc simulation, *Methods in Ecology and Evolution*, 2022, 13 (2), 77–90, doi:10.1111/2041-210X.13727.

[J34] **Höhna S**, Landis MJ and Huelsenbeck JP: Parallel power posterior analyses for fast computation of Bayes factors in phylogenetics, *PeerJ*, 2021, 9:e12438, doi:10.7717/peerj.12438.

[J35] Orsi WD, Magritsch T, Vargas S, Coskun ÖK, Vuillemin A, **Höhna S**, Wörheide G, D'Hondt S, Shapiro BJ, Carini P: Genome evolution in bacteria isolated from million-year-old subseafloor sediment, *mBio*, 2021, 12:e01150-21, doi:10.1128/mBio.01150-21.

[J36] Magee AF, **Höhna S**, Vasylyeva TI, Leaché AD, and Minin VN: Locally adaptive Bayesian birth-death model successfully detects slow and rapid rate shifts, *PLoS Comp Biol*, 2020, 16(10): e1007999, doi:10.1371/journal.pcbi.1007999.

[J37] Catalán A, Briscoe A and **Höhna S**: Drift and directional selection are the evolutionary forces driving gene expression divergence in eye and brain tissue of Heliconius butterflies, *Genetics*, 2019, 213 (2), 581–594, doi:10.1534/genetics.119.302493.

[J38] Freyman WA and **Höhna S**: Stochastic Character Mapping of State-Dependent Diversification Reveals the Tempo of Evolutionary Decline in Self-Compatible Onagraceae Lineages, *Systematic Biology*, 2019, 68 (3), 505–519, doi:10.1093/sysbio/syy078.

[J39] Silvestro D, Tejedor MF, Serrano-Serrano ML, Loiseau O, Rossier V, Rolland J, Zizka A, **Höhna S**, Antonelli A and Salamin N: Early Arrival and Climatically-Linked Geographic Expansion of New World Monkeys from Tiny African Ancestors, *Systematic Biology*, 2019, 68 (1), 78–92, doi:10.1093/sysbio/syy046.

[J40] Condamine FL, Rolland J, **Höhna S**, Sperling FAH and Sanmartín I: Testing the Role of the Red Queen and Court Jester as Drivers of the Macroevolution of Apollo Butterflies, *Systematic Biology*, 2018, 67 (6), 940–964, doi:10.1093/sysbio/syy009.

[J41] Martin CH and **Höhna S**: New evidence for the recent divergence of Devil's Hole pupfish and the plausibility of elevated mutation rates in endangered taxa, *Molecular Ecology*, 2018, 27 (4), 831–838, doi:10.1111/mec.14404.

[J42] **Höhna S**, Coghill LM, Mount G, Thomson R and Brown JM: P^3: Phylogenetic Posterior Prediction in RevBayes, *Molecular Biology and Evolution*, 2018, 35 (4), 1028–1034, doi:10.1093/molbev/msx286.

[J43] Freyman WA and **Höhna S**: Cladogenetic and Anagenetic Models of Chromosome Number Evolution: a Bayesian Model Averaging Approach, *Systematic Biology*, 2018, 67 (2), 195–215 doi:10.1093/sysbio/syx065.

[J44] Martin CH, **Höhna S**, Crawford JE, Turner BJ, Richards EJ and Simons LH: The complex effects of demographic history on the estimation of substitution rate: concatenated analysis results in no more than 2-fold overestimation, *Proceedings of the Royal Society B*, 2017, 284: 20170537, doi:10.1098/rspb.2017.0537.

[J45] **Höhna S**, Landis MJ and Heath TA: Phylogenetic Inference using RevBayes, *Current Protocols in Bioinformatics*, 2017, 57:6.16.1-6.16.34, doi:10.1002/cpbi.22.

[J46] Moore BR, **Höhna S**, May MR, Rannala B and Huelsenbeck JP: Critically evaluating the theory and performance of Bayesian analysis of macroevolutionary mixtures. *Proceedings of the National Academy of Sciences*, 2016, 113, 9569–9574, doi:10.1073/pnas.1518659113.

[J47] **Höhna S**, Landis MJ, Heath TA, Boussau B, Moore BR, Lartillot N, Huelsenbeck JP and Ronquist F: RevBayes: Bayesian Phylogenetic Inference Using Graphical Models and an Interactive Model-Specification Language, *Systematic Biology*, 2016, 65 (4), 726–736 doi:10.1093/sysbio/syw021.

[J48] May MR, **Höhna S** and Moore BR: A Bayesian Approach for Detecting the Impact of Mass-Extinction Events on Molecular Phylogenies When Rates of Lineage Diversification May Vary, *Methods in Ecology and Evolution*, 2016, doi:10.1111/2041-210X.12563.

[J49] Conroy CR, Patton JL, Lim M, Phuong M, Parmenter B and **Höhna S**: Following the rivers: historical reconstruction of California voles (Microtus californicus, Muridae, Rodentia) in the deserts of eastern California, *Biological Journal of the Linnean Society*, 2016, 119, 80-98 doi:10.1111/bij.12808.

[J50] **Höhna S**, May MR and Moore BR: `TESS`: An R package for efficiently simulating phylogenetic trees and performing Bayesian inference of lineage diversification rates, *Bioinformatics*, 2016, 32 (5), 789–791. doi:10.1093/bioinformatics/btv651

[J51] Janzen T, **Höhna S**, Etienne RS: Approximate Bayesian Computation of diversification rates from molecular phylogenies: introducing a new efficient summary statistic, the nLTT, *Methods in Ecology and Evolution*, 2015, 61 (1), 1–11. doi:10.1111/2041-210X.12350

[J52] **Höhna S**: The time-dependent reconstructed evolutionary process with a key-role for mass-extinction events, *Journal of Theoretical Biology* 2015, 380, 321–331. doi:10.1016/j.jtbi.2015.06.005

[J53] **Höhna S**, Heath TA, Boussau B, Landis MJ, Ronquist F and Huelsenbeck JP: Graphical Model Representation in Phylogenetics, *Systematic Biology*, 2014, 63 (5). 753–771. doi:10.1093/sysbio/syu039

[J54] **Höhna S**: Likelihood Inference of non-constant Diversification Rates with Incomplete Taxon Sampling, *PLoS One*, 2014; 9(1): e84184, doi:10.1371/journal.pone.0084184

[J55] **Höhna S**: Fast simulation of reconstructed phylogenies under global time-dependent birth-death processes, *Bioinformatics*, 2013, 29 (11), 1367–1374, doi:10.1093/bioinformatics/btt153

[J56] Ronquist F, Teslenko M, van der Mark P, Ayres DL, Darling A, **Höhna S**, Larget B, Liu L, Suchard MA and Huelsenbeck JP: `MrBayes` 3.2: Efficient Bayesian MCMC Inference and Model Choice Across a Large Model Space, *Systematic Biology*, 2012, 61 (3): 539-542. doi:10.1093/sysbio/sys029

[J57] **Höhna S** and Drummond AJ: Guided Tree Topology Proposals for Bayesian Phylogenetic Inference, *Systematic Biology*, 2012, 61 (1), 1-11. doi:10.1093/sysbio/syr074

[J58] Alström P, **Höhna S**, Gelang M, Ericson P and Olsson U: Non-monophyly and intricate morphological evolution within the avian family Cettiidae revealed by multilocus analysis of a taxonomically densely sampled dataset, *BMC Evolutionary Biology*, 11 (1), 352, doi:10.1186/1471-2148-11-352

[J59] **Höhna S**, Stadler T, Ronquist F, Britton T: Inferring speciation and extinction rates under different sampling schemes, *Molecular Biology and Evolution*, 2011, 28 (9), 2577-2589, doi:10.1093/molbev/msr095

Publications in conference proceedings:

[P1] **Höhna S**, Defoin-Platel M, Drummond AJ: Clock-constrained Tree Proposal Operators in Bayesian Phylogenetic Inference, *8th IEEE International Conference on BioInformatics and BioEngineering*, Athens, September 2008

Scientific Presentations

Contributed Scientific Presentations at conferences and workshops:

Oct 2025	TaxonOmics (invited), online, Germany
May 2025	MCEB, Granada, Spain
Jul 2024	Evolution Meetings, Montreal, Canada
Apr 2024	ProbGen (Invited Session Chair), Vienna, Austria
Jun 2023	MCEB (Keynote), Corsica, France
Oct 2022	LMU Munich (GeoBio-Center Symposium), Germany
Aug 2021	PhyBae (Bavarian Phylogenetic Meeting), Munich, Germany
Jan 2020	SSB Meetings, Gainesville, USA
May 2019	Phyloseminar (webcast)
Aug 2018	Evolution Meetings, Montpellier, France
Jul 2018	SMBE Meetings, Oklahoma, Japan
Jan 2017	SSB Meetings, Baton Rouge, USA
Jun 2015	Evolution Meetings, Sao Paolo, Brazil
Jun 2013	Evolution Meetings (Ernst Mayr Award Symposium), Snowbird, USA
Jun 2012	SMBE Meetings, Dublin, Ireland
Feb 2012	Phyloseminar (webcast)
Aug 2011	ESEB Meetings, Tübingen, Germany
Jun 2011	iEvoBio, Oklahoma, USA
Jun 2011	Evolution Meetings, Oklahoma, USA
Aug 2010	IMS, Gothenburg
Jul 2010	SMBE Meetings, Lyon, France
Jun 2010	Evolution Meetings (Ernst Mayr Award Symposium), Portland
Jan 2010	Workshop on Molecular Evolution, Cesky Krumlov
Oct 2008	BIBE, Athens
Mar 2008	NZCRSC, Christchurch

Invited Scientific Seminars at Universities:

Nov 2022	LMU Munich (Botanical Garden), Germany
Sep 2022	Bielefeld University, Germany
Apr 2022	Universidad de San Carlos de Guatemala, Guatemala
Nos 2021	Julius-Maximilians-Universität Würzburg, Germany
Dec 2019	Natural History Museum, Oslo
Oct 2019	Swedish Museum of Natural History, Stockholm
May 2019	23andMe, Mountain View
Dec 2018	Université de Montpellier, Montpellier
Jul 2016	EES, Ludwig-Maximilians-Universität, Munich
Apr 2016	EEB, University of Rochester
Mar 2016	EDG, University of British Columbia
Nov 2015	LBBE, Universite Claude Bernard Lyon 1
Oct 2015	Department of Biological Sciences, Louisiana State University
Sep 2015	CTEG, UC Berkeley
Nov 2014	Biostatistics, UC Davis
Oct 2014	Center for Population Biology, UC Davis
Oct 2014	Neyman Seminar (Statistics), UC Berkeley
Sep 2014	CTEG, UC Berkeley
Mar 2014	Evolution, ETH Zurich
Jun 2013	LBBE, Universite Claude Bernard Lyon 1
Feb 2013	EBC, Uppsala University
Sep 2012	Department of Mathematics, Stockholm University
Aug 2011	RevBayes Workshop, UC Berkeley
Aug 2011	Bayesian Phylogenetic Inference Symposium, UC Berkeley
May 2011	Stockholm-Uppsala Mathematical Statistics Symposium, Uppsala
May 2010	Department of Mathematics, Stockholm University
Apr 2009	Department of Mathematics, Stockholm University
Mar 2009	Department of Computer Science, University of Auckland

Posters presented:

June 2012	Mathematical and Computational Evolutionary Biology, Montpellier
June 2009	ISMB/ECCB, Stockholm

Research Grants, Scholarships and Awards

2027	20,000€	DFG Conference Organization: "Mathematical and Computational Evolutionary Biology Meeting", Gepris
2026 – 2027	51,118€	DFG Sachbeihilfe: "Efficient Species Delimitation using Sequential Bayesian Inference", Gepris
2026 – 2029	172,250 €	DFG Sachbeihilfe: "Macroevolutionary Drivers if Trait Evolution", Gepris
2026 – 2029	199,150 €	DFG SPP-2349: "Genomic Basis of Evolutionary Innovations (GEvol)" (joint-main PI, with Dr. Ana Catalán), Total project funding: 534,657 €, Gepris
2025 – 2026	100,000 €	LMU Investment Fund
2024 – 2028	870,800 €	ATRAE: "Species Delimitation from Genomic Polymorphism Data" (declined)
2023 – 2026	151,050 €	DFG SPP-2349: "Genomic Basis of Evolutionary Innovations (GEvol)" (joint-main PI, with Dr. Ana Catalán), Total project funding: 505,776 €, Gepris
2022 – 2027	1.5 Mio €	**ERC Starting Grant**
2020 – 2023	145,500 €	DFG SPP-1991 TaxonOMICS (with Dr. Ana Catalán), Gepris
2019 – 2023	1.4 Mio €	**Emmy Noether Program, DFG**, Gepris
2019 – 2020	50,000 €	LMU Junior Research Fund
2019 – 2020	50,000 €	KONWIHR-Förderprogram (with Prof. Gert Wörheide)
2018	4,600 €	BayLat Research Collaboration Initiation Grant
2014 – 2017	219,000 $	Miller Fellowship, Miller Institute for Basic Research in Science, Berkeley
2015 – 2016	10,000 $	XSEDE Cluster Allocation
2013	20,000 SEK	The G.S. Magnusson Foundation, Travel Award, Swedish Royal Academy of Sciences
2012	2,400 €	Erasmus Study Scholarship
2012	20,000 SEK	The G.S. Magnusson Foundation, Travel Award, Swedish Royal Academy of Sciences
2011	20,000 SEK	The G.S. Magnusson Foundation, Travel Award, Swedish Royal Academy of Sciences
2010	20,000 SEK	The G.S. Magnusson Foundation, Travel Award, Swedish Royal Academy of Sciences
2010	2,000 €	Graduate Student Travel Award, Society of Molecular Biology and Evolution
2008	3,000 NZD	Faculty of Science Masters Award, University of Auckland, Auckland
2007	-	New Zealand Programming Champion

Miscellaneous

Local Host: Applied Probability Society Conference, Stockholm, 2011.

Handling Editor: Systematic Biology (since 2024)

Editorial Board: Systematic Biology (since 2020)

Reviewer for (Journals): Bayesian Analysis, BMC Evolutionary Biology, Bioinformatics, Evolution, Evolution Letters, Journal of Theoretical Biology, Methods in Ecology and Evolution, Molecular Biology and Evolution, Nature, Nature Ecology & Evolution, Nature Communications, New Phytologist, PeerJ, PNAS, Proceedings of the Royal Society B, Systematic Biology

Reviewer for (Grants): Austrian Science Fund (FWF), European Research Council (ERC), Research Council KU Leuven, Society for Systematic Biology (SSB), Swiss National Science Foundation (SNF)